The European Variation Archive: a FAIR resource of genomic variation for all species

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EMBL-EBI

European **Variation** Archive

An open-access data sharing and analysis platform for all types of genetic variation data from any species.

Responsible for accessioning of variants from **non-human** type species

What is the European Variation Archive?

Variant data repository | **Deposition database**

- **Submission-based** service for variation in VCF
- Studies permanently archived and **accessioned for publication**
- Data validated, normalized and **annotated** via Variant Effect Predictor
- As of June 2022, the EVA contains
  - 2,170 short variation studies
  - 204 structural variation studies
  - 3.8+ billion browsable variants

How to submit data?

Prepare

- Prepare valid VCF file(s) and validate with EVA VCF validation suite.
- Complete a metadata template describing the samples and analyses in your study.

Contact

Contact eva-helpdesk@ebi.ac.uk to request a submission.

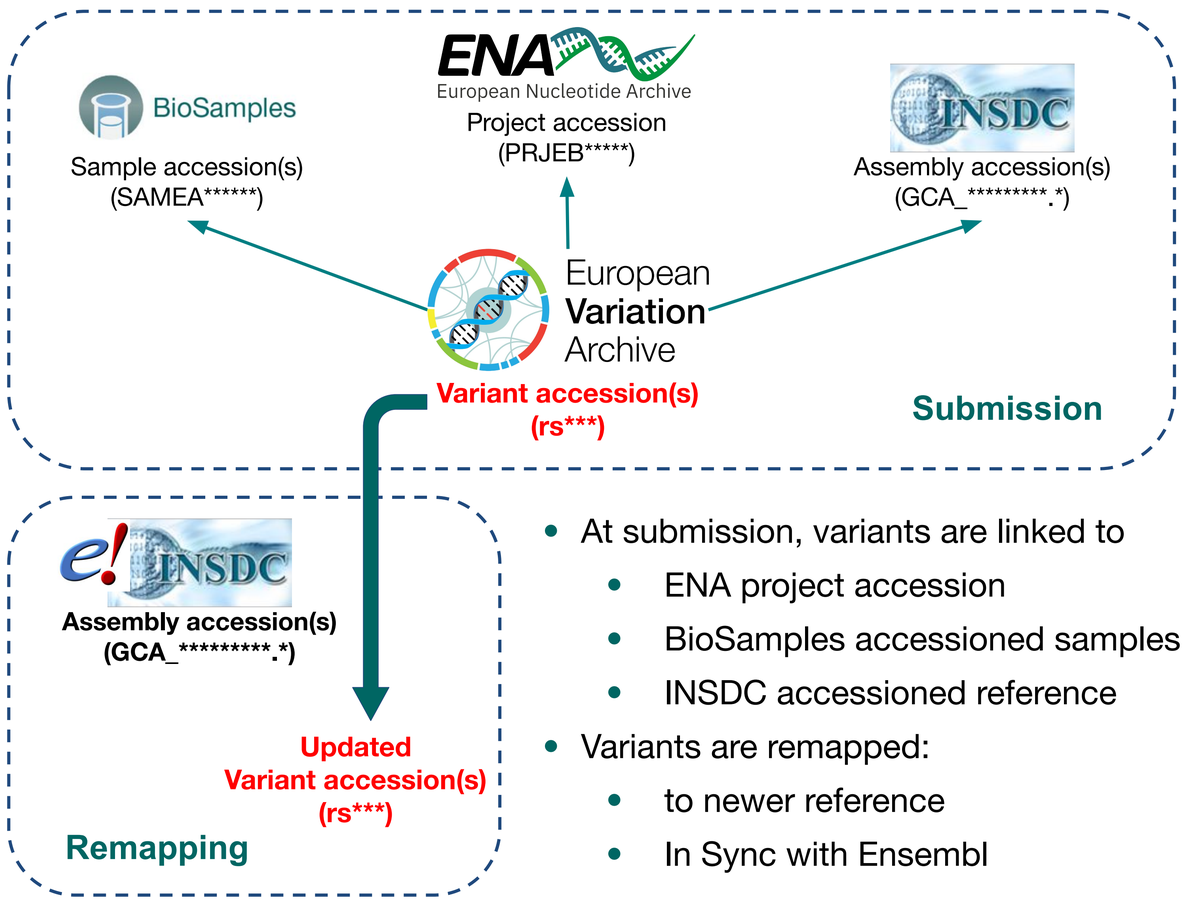
Submit

Upload your VCF file(s), metadata template and any associated data file(s) to your private FTP location.

Receive

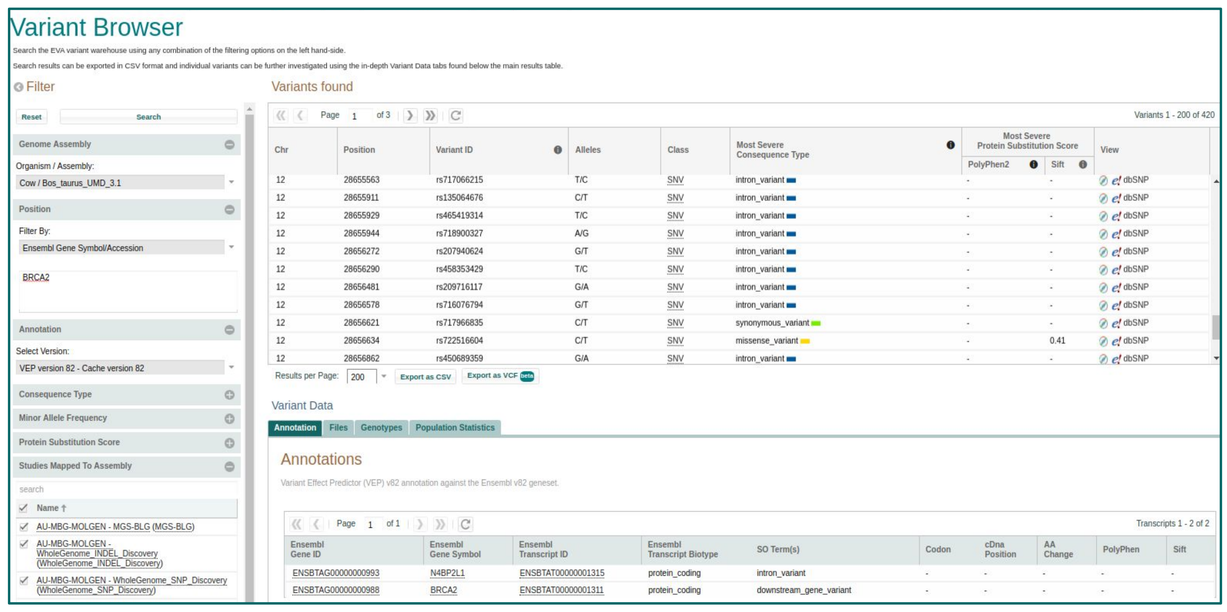
Accession numbers will be sent via email to the submitter upon successful archival of the deposited data.

Findable, cross-linked, long term, up-to-date accessions



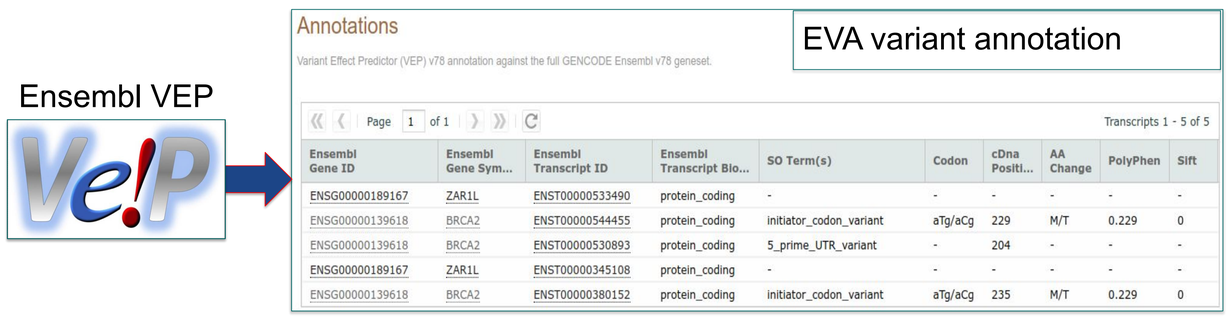
- At submission, variants are linked to
  - ENA project accession
  - BioSamples accessioned samples
  - INSDC accessioned reference
- Variants are remapped:
  - to newer reference
  - In Sync with Ensembl

Variant data consumption



- **Variant Browser**: Filters on the left sidebar, variants found on the right such as ID, alleles, mutation type, PolyPhen and SIFT scores etc
- Variant information can be also retrieved programmatically using our REST **API.** Documentation and examples in "API" section on the EVA website

Ensembl VEP



- Ensembl's variant effect predictor (VEP) used to determine effect of variants held within EVA
- Potential in adding clinical significance to variants

EVA with ELIXIR

Human Copy Number Variation

- Improve representation of Structural variants in VCF 4.4
- VCF 4.4 Release candidate is here. Check it out at

Plant Sciences

- Part of the FAIR-ification of Plant Genotyping Data and its linking to Phenotyping using ELIXIR Platforms (FONDUE) project
- Improvement of submission practice for plant variants
- Recommendation for formatting VCF